

Risk and Return

WHAT IS RISK?

- **Risk** is a measure of the uncertainty surrounding the return that an investment will earn. Investments whose returns are more uncertain are generally riskier.

WHAT IS RETURN?

- The **total rate of return** (sometimes called the holding period return) is the total gain or loss experienced on an investment over a given period expressed as a percentage of the investment's value at the beginning of the period.
- Mathematically, an investment's total return is the sum of any cash distributions (e.g., dividends or interest payments) plus the change in the investment's price, divided by the beginning-of period price

$$r_t = \frac{P_t - P_{t-1} + C_t}{P_{t-1}} \quad (8.1)$$

where

r_t = total return during period t

P_t = price of asset at time t

P_{t-1} = price of asset at time $t-1$

C_t = cash flow received from the asset during period t

WHAT IS RETURN?

- Robin wishes to determine the return on two stocks she owned during 2019, McDonald's and Walmart. At the beginning of the year, McDonald's stock traded for \$170.84 per share, and Walmart stock was valued at \$90.69. During the year, McDonald's paid \$4.73 per share in dividends, and Walmart shareholders received dividends of \$2.12 per share. At the end of the year, McDonald's stock was worth \$196.38, and Walmart stock sold for \$117.80.

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$$r_t = \frac{P_t - P_{t-1} + C_t}{P_{t-1}}$$

McDonald's: $(\$196.38 - \$170.84 + \$4.73) \div \$170.84 = 17.7\%$

Walmart: $(\$117.80 - \$90.69 + \$2.12) \div \$90.69 = 32.2\%$

where

r_t = total return during period t

P_t = price of asset at time t

P_{t-1} = price of asset at time $t-1$

C_t = cash flow received from the asset during period t

Defining Risk

- *The variability of returns from those that are expected.*

What rate of return do you expect on your investment (savings) this year?

What rate will you actually earn?

Does it matter if it is a bank CD or a share of stock?

Probability Distributions to Measure Risk

- For risky securities, the actual rate of return can be viewed as a random variable subject to a **probability distribution**.
- Probability distribution: A set of possible values that a random variable can assume and their associated probabilities of occurrence.
- This probability distribution can be summarized in terms of two parameters of the distribution: (1) the *expected return* and (2) the *standard deviation*.

Expected Return and Standard Deviation

- Expected return: The weighted average of possible returns, with the weights being the probabilities of occurrence.

The expected return, $\bar{R}$, is

$$\bar{R} = \sum_{i=1}^n (R_i)(P_i)$$

- $\bar{R}$ is the expected return for the asset,
- R_i is the return for the i^{th} possibility,
- P_i is the probability of that return occurring,
- n is the total number of possibilities.

$$\bar{r} = \sum_{j=1}^n r_j \times Pr_j$$

Expected Return and Stan

where

r_j = return for the j th outcome

Pr_j = probability of occurrence of the j th outcome

n = number of outcomes considered

Stock BW		
R_j	P_j	$(R_j)(P_j)$
-0.15	0.10	-0.015
-0.03	0.20	-0.006
0.09	0.40	0.036
0.21	0.20	0.042
0.33	0.10	0.033
Sum	1.00	0.090

The
expected
return, $\bar{R}$,
for Stock
BW is .09
or 9%

Expected Return and Standard Deviation

- We need a measure of the dispersion, or variability, around our expected return. The conventional measure of dispersion is the **standard deviation**.
- The greater the standard deviation of returns, the greater the variability of returns, and the greater the risk of the investment. The standard deviation, σ , can be expressed mathematically as

$$\sigma = \sqrt{\sum_{j=1}^n [(r_j - \bar{r})^2 \times Pr_j]}$$

$$\sigma = \sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 (P_i)}$$

- It is the square root of variance. Note, this is for a *discrete distribution*.

$$\sigma^2 = \sum_{j=1}^n [(r_j - \bar{r})^2 \times Pr_j]$$

TABLE 8.3 Average Returns for Norman Company Assets A and B

Possible outcomes	Probability Pr_j	Returns r_j	$Pr_j \times r_j$
Asset A			
Pessimistic	0.25	13%	3.25%
Most likely	0.50	15	7.50
Optimistic	<u>0.25</u>	17	<u>4.25</u>
Total	<u>1.00</u>		Average return <u>15.00%</u>
Asset B			
Pessimistic	0.25	7%	1.75%
Most likely	0.50	15	7.50
Optimistic	<u>0.25</u>	23	<u>5.75</u>
Total	<u>1.00</u>		Average return <u>15.00%</u>

$$\sigma = \sqrt{\sum_{j=1}^n [(r_j - \bar{r})^2 \times Pr_j]}$$

TABLE 8.4 Calculating the Standard Deviation of the Returns for Norman Company Assets A and B

Asset A j	Squared deviation $(r_j - \bar{r})^2$	Probability Pr_j	$(r_j - \bar{r})^2 \times Pr_j$
1	$(13\% - 15\%)^2 = 4\%^2$	0.25	$4\%^2 \times 0.25 = 1\%^2$
2	$(15\% - 15\%)^2 = 0\%^2$	0.50	$0\%^2 \times 0.25 = 0\%^2$
3	$(17\% - 15\%)^2 = 4\%^2$	0.25	$4\%^2 \times 0.25 = 1\%^2$
$\sum_{j=1}^3 (r_j - \bar{r})^2 \times Pr_j = 1\%^2 + 0\%^2 + 1\%^2 = 2\%^2$			
$\sigma = \sqrt{\sum_{j=1}^3 (r_j - \bar{r})^2 \times Pr_j} = \sqrt{2\%^2} = \underline{\underline{1.41\%}}$			
Asset B			
1	$(7\% - 15\%)^2 = 64\%^2$	0.25	$64\%^2 \times 0.25 = 16\%^2$
2	$(15\% - 15\%)^2 = 0\%^2$	0.50	$0\%^2 \times 0.25 = 0\%^2$
3	$(23\% - 15\%)^2 = 64\%^2$	0.25	$64\%^2 \times 0.25 = 16\%^2$
$\sum_{j=1}^3 (r_j - \bar{r})^2 \times Pr_j = 16\%^2 + 0\%^2 + 16\%^2 = 32\%^2$			
$\sigma = \sqrt{\sum_{j=1}^3 (r_j - \bar{r})^2 \times Pr_j} = \sqrt{32\%^2} = \underline{\underline{5.66\%}}$			

POSSIBLE RETURN, R_i	PROBABILITY OF OCCURRENCE, P_i	EXPECTED RETURN ($\bar{R}$) CALCULATION	VARIANCE (σ^2) CALCULATION
		$(R_i)(P_i)$	$(R_i - \bar{R})^2(P_i)$
-0.10	0.05	-0.005	$(-0.10 - 0.09)^2(0.05)$
-0.02	0.10	-0.002	$(-0.02 - 0.09)^2(0.10)$
0.04	0.20	0.008	$(0.04 - 0.09)^2(0.20)$
0.09	0.30	0.027	$(0.09 - 0.09)^2(0.30)$
0.14	0.20	0.028	$(0.14 - 0.09)^2(0.20)$
0.20	0.10	0.020	$(0.20 - 0.09)^2(0.10)$
0.28	<u>0.05</u>	<u>0.014</u>	<u>$(0.28 - 0.09)^2(0.05)$</u>
	$\Sigma = 1.00$	$\Sigma = 0.090 = \bar{R}$	$\Sigma = 0.00703 = \sigma^2$
		Standard deviation = $(0.00703)^{0.5} = 0.0838 = \sigma$	

Coefficient of Variation

- The **coefficient of variation**, CV , is a measure of relative dispersion that is useful in comparing assets with different standard deviations and different average, or expected, returns.

$$CV = \frac{\sigma}{\bar{r}}$$

- A higher coefficient of variation means that an investment has more volatility relative to its expected return. Because investors prefer higher returns and less risk, one might intuitively expect investors to gravitate toward investments with a low coefficient of variation

- Marilyn Ansbro is reviewing stocks for inclusion in her investment portfolio. The stock she wishes to analyze is Danhaus Industries Inc. (DII), a diversified manufacturer of pet products. One of her key concerns is risk; as a rule, she will invest only in stocks with a coefficient of variation below 0.75. She has gathered price and dividend data (shown in the accompanying table) for DII over the past three years, and she plans to calculate DII's coefficient of variation using this admittedly limited historical sample.

Year	Stock Price		Dividend paid
	Beginning	Ending	
1	\$35.00	\$36.50	\$3.50
2	36.50	34.50	3.50
3	34.50	35.00	4.00

Substituting the price and dividend data for each year into Equation 8.1, we calculate the return on DII stock in each year.

Year	Returns
1	$(\$36.50 - \$35.00 + \$3.50) \div \$35.00 = 14.3\%$
2	$(\$34.50 - \$36.50 + \$3.50) \div \$36.50 = 4.1\%$
3	$(\$35.00 - \$34.50 + \$4.00) \div \$34.50 = 13.0\%$
$\bar{r} = (14.3\% + 4.1\% + 13.0\%) \div 3 = 10.5\%$	

Sub

Substituting the average return and annual returns into Equation 8.4a, we get the standard deviation, σ :

$$\begin{aligned}\sigma &= \sqrt{[(14.3\% - 10.5\%)^2 + (4.1\% - 10.5\%)^2 + (13.0\% - 10.5\%)^2] \div (3 - 1)} \\ &= \sqrt{(14.44\% + 40.96\% + 6.25\%) \div 2} = \sqrt{30.825\%} = 5.6\%\end{aligned}$$

Finally, substituting the standard deviation of returns and the average return into Equation 8.5, we get the coefficient of variation, CV:

$$CV = 5.6\% \div 10.5\% = 0.53$$

Because the coefficient of variation of returns on the DII stock over this period of 0.53 is well below Marilyn's maximum coefficient of variation of 0.75, she concludes that the DII stock would be an acceptable investment.

RISK PREFERENCES

- **Certainty Equivalent (CE)** is the amount of cash someone would require with certainty at a point in time to make the individual indifferent between that certain amount and an amount expected to be received with risk at the same point in time.
- Economists use three categories to describe how investors respond to risk.
- investors who are **risk seeking** prefer investments with higher risk, so much so that they may choose investments with very low expected returns for the thrill of taking extra risk. Although most individuals do not exhibit this behavior most of the time, it is not difficult to find examples of risk-seeking behavior, particularly in the realm of gambling.
- Investors who are **risk neutral** choose investments based solely on their expected returns, disregarding the risks. When choosing between two investments, *risk-neutral investors will always select the investment with the higher expected return regardless of its risk.*
- Investors who are **risk averse** prefer less risky over more risky investments, holding the expected rate of return fixed. A risk-averse investor who believes that two different investments have the same expected return will choose the investment whose returns are more certain.

Risk Attitude Example

- You have the choice between (1) a guaranteed dollar reward or (2) a coin-flip gamble of \$100,000 (50% chance) or \$0 (50% chance). The expected value of the gamble is \$50,000.
 - *Mary* requires a guaranteed \$25,000, or more, to call off the gamble.
 - *Rashid* is just as happy to take \$50,000 or take the risky gamble.
 - *Shan* requires at least \$52,000 to call off the gamble.

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 - *Mary* requires a guaranteed \$25,000, or more, to call off the gamble.
 - *Rashid* is just as happy to take \$50,000 or take the risky gamble.
 - *Shan* requires at least \$52,000 to call off the gamble.

Mary shows “risk aversion” because her “certainty equivalent” $<$ the expected value of the gamble.

Rashid exhibits “risk indifference” because her “certainty equivalent” equals the expected value of the gamble.

Shan reveals a “risk preference” because her “certainty equivalent” $>$ the expected value of the gamble.

Risk and Return in a Portfolio

- Investors want to create an **efficient portfolio**, one that provides the maximum return for a given level of risk. We therefore need a way to measure the return and the standard deviation of a portfolio of assets.
- Portfolio: A combination of two or more securities or assets.

PORTFOLIO RETURN

- The expected return of a portfolio is simply a weighted average of the expected returns of the securities constituting that portfolio. The weights are equal to the proportion of total funds invested in each security (the weights must sum to 100 percent)

$$r_p = (w_1 \times r_1) + (w_2 \times r_2) + \dots (w_n \times r_n) = \sum_{j=1}^n w_j \times r_j \quad (8.)$$

Where

w_j = percentage of the portfolio's total dollar value invested in asset j

r_j = return on asset j

$$\bar{R}_P = \sum_{j=1}^m (W_j)(\bar{R}_j)$$

$$r_p = (w_1 \times r_1) + (w_2 \times r_2) + \dots (w_n \times r_n) = \sum_{j=1}^n w_j \times r_j \quad (8)$$

PORTFOLIO RETUF

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w_j = percentage of the portfolio's total dollar value invested in asset j

r_j = return on asset j

- James purchases 10 shares of JPMorgan at \$90 per share, so his total investment in JPMorgan is \$900. He also buys 27 shares of Medtronic at \$100 per share, so the total investment in Medtronic is \$2,700. Combining these two holdings, James's total portfolio is worth \$3,600. Of the total, 25% is invested in JPMorgan ($900 / 3,600$), and 75% is invested in Medtronic ($+2,700 / +3,600$). Thus, $w_1 = 0.25$, $w_2 = 0.75$, and $w_1 + w_2 = 1.0$

$$\bar{r}_p = (0.25 \times 3.4\%) + (0.75 \times 2.1\%) = 2.4\%$$

Portfolio Risk and the Importance of Covariance

- *Covariance* is a statistical measure of the degree to which two variables (e.g., securities' returns) move together.
- Positive covariance shows that, on average, the two variables move together. Negative covariance suggests that, on average, the two variables move in opposite directions.
- Zero covariance means that the two variables show no tendency to vary together in either a positive or negative linear fashion.

What is Covariance?

- The *covariance* of the possible returns of two securities is a measure of the extent to which they are expected to vary together rather than independently of each other.

$$\sigma_{j,k} = r_{j,k} \sigma_j \sigma_k$$

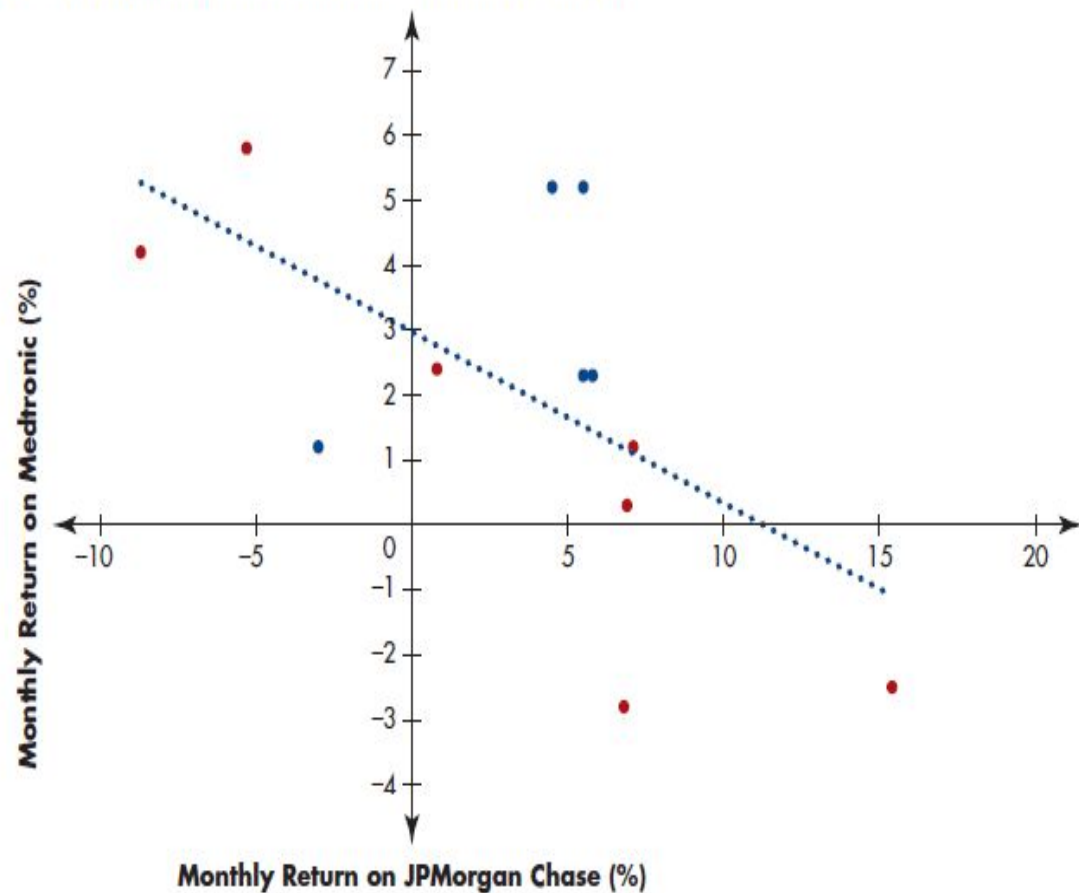
- where $r_{j,k}$ is the expected correlation coefficient between possible returns for securities j and k , σ_j is the standard deviation for security j , and σ_k is the standard deviation for security k .

Correlation

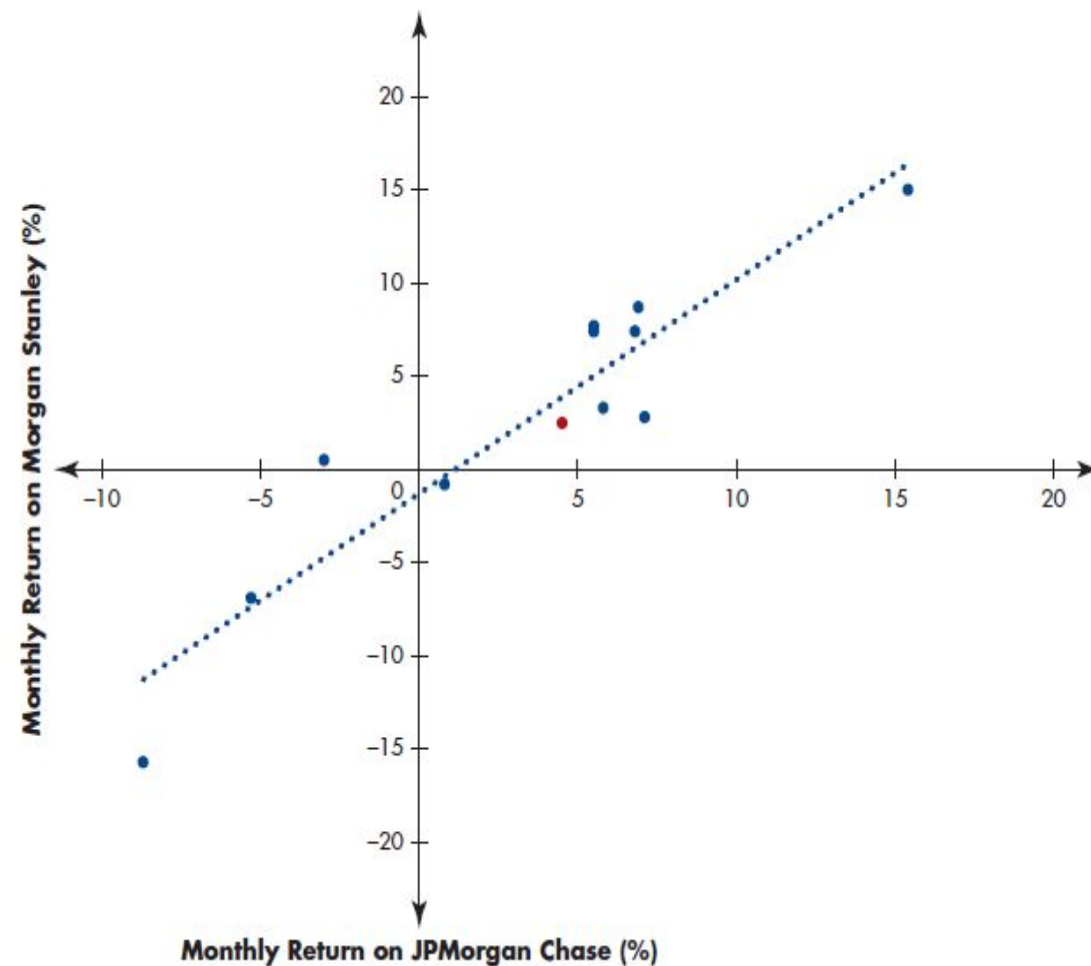
- Correlation is a statistical measure of the relationship between any two series of numbers. The numbers may represent data of any kind, from investment returns to test scores.
- The correlation coefficient measures the degree of correlation and takes values between -1 and +1.
- For stocks that are perfectly positively correlated, the correlation coefficient equals +1, and a plot of the returns on these stocks would show that they lie exactly along an upward sloping line,
- For stocks that are perfectly negatively correlated, the correlation coefficient equals -1.0, and a plot of the returns on those stocks would show that they fall exactly on a downward sloping line.
- For stocks that are **uncorrelated**, the correlation coefficient equals zero, which means there is no relation between the returns on the two stocks.

FIGURE 8.4

Part A: Monthly Returns on JPMorgan Chase and Medtronic



Part B: Monthly Returns on JPMorgan Chase and Morgan Stanley



How to calculate correlation coefficient

To calculate the correlation coefficient for a pair of stocks, we need historical returns for each stock.

- The first step is to calculate the average return and standard deviation of returns for each stock.
- Second, for each of the n historical periods (e.g., each month or year), we calculate the difference between each stock's return and its average return and divide that by the standard deviation. This is the "standardized" return.
- Third, in each period we multiply the standardized return for one stock times the standardized return for the other.
- Fourth, we add up these terms over time and divide by $n - 1$.
Mathematically, the correlation coefficient between two stocks i and j , ρ_{ij} , calculated using n historical observations.

$$\rho_{ij} = \sum_{t=1}^n \left[\frac{(r_{t,i} - \bar{r}_i)}{\sigma_i} \times \frac{(r_{t,j} - \bar{r}_j)}{\sigma_j} \right] \div (n - 1) \quad (8.8)$$

$$\rho_{ij} = \frac{\frac{(r_{1i} - \bar{r}_i)}{\sigma_i} \times \frac{(r_{1j} - \bar{r}_j)}{\sigma_j} + \frac{(r_{2i} - \bar{r}_i)}{\sigma_i} \times \frac{(r_{2j} - \bar{r}_j)}{\sigma_j} + \dots + \frac{(r_{ni} - \bar{r}_i)}{\sigma_i} \times \frac{(r_{nj} - \bar{r}_j)}{\sigma_j}}{(n - 1)}$$

TABLE 8.7

Correlati

$$\rho_{ij} = \frac{\frac{(r_{1i} - \bar{r}_i)}{\sigma_i} \times \frac{(r_{1j} - \bar{r}_j)}{\sigma_j} + \frac{(r_{2i} - \bar{r}_i)}{\sigma_i} \times \frac{(r_{2j} - \bar{r}_j)}{\sigma_j} + \dots + \frac{(r_{ni} - \bar{r}_i)}{\sigma_i} \times \frac{(r_{nj} - \bar{r}_j)}{\sigma_j}}{(n - 1)}$$

	(1)	(2)	(3)	(4)	(5)
Month _t	JPMorgan return, $r_{t,JPM}$	$(r_{t,JPM} - \bar{r}_{JPM}) \div \sigma_{JPM}$	Medtronic return, $r_{t,MDT}$	$(r_{t,MDT} - \bar{r}_{MDT}) \div \sigma_{MDT}$	(2) × (4)
Jan	6.8%	$(6.8\% - 3.4\%) \div 6.5\% = 0.523$	-2.8%	$(-2.8\% - 2.1\%) \div 2.8\% = -1.750$	-0.915
Feb	0.8	$(0.8\% - 3.4\%) \div 6.5\% = -0.400$	2.4	$(2.4\% - 2.1\%) \div 2.8\% = 0.107$	-0.043
Mar	-3.0	$(-3.0\% - 3.4\%) \div 6.5\% = -0.985$	1.2	$(1.2\% - 2.1\%) \div 2.8\% = -0.321$	0.316
Apr	15.4	$(15.4\% - 3.4\%) \div 6.5\% = 1.846$	-2.5	$(-2.5\% - 2.1\%) \div 2.8\% = -1.643$	-3.033
May	-8.7	$(-8.7\% - 3.4\%) \div 6.5\% = -1.862$	4.2	$(4.2\% - 2.1\%) \div 2.8\% = 0.750$	-1.396
Jun	5.5	$(5.5\% - 3.4\%) \div 6.5\% = 0.323$	5.2	$(5.2\% - 2.1\%) \div 2.8\% = 1.107$	0.358
Jul	4.5	$(4.5\% - 3.4\%) \div 6.5\% = 0.169$	5.2	$(5.2\% - 2.1\%) \div 2.8\% = 1.107$	0.187
Aug	-5.3	$(-5.3\% - 3.4\%) \div 6.5\% = -1.338$	5.8	$(5.8\% - 2.1\%) \div 2.8\% = 1.321$	-1.769
Sep	7.1	$(7.1\% - 3.4\%) \div 6.5\% = 0.569$	1.2	$(1.2\% - 2.1\%) \div 2.8\% = -0.321$	-0.183
Oct	6.9	$(6.9\% - 3.4\%) \div 6.5\% = 0.538$	0.3	$(0.3\% - 2.1\%) \div 2.8\% = -0.643$	-0.346
Nov	5.5	$(5.5\% - 3.4\%) \div 6.5\% = 0.323$	2.3	$(2.3\% - 2.1\%) \div 2.8\% = 0.071$	0.023
Dec	5.8	$(5.8\% - 3.4\%) \div 6.5\% = 0.369$	2.3	$(2.3\% - 2.1\%) \div 2.8\% = 0.071$	0.026
Average return, $\bar{r}$	3.4		2.1		
Standard deviation, σ	6.5		2.8		
				Sum	-6.78
				Correlation Coefficient [Sum ÷ (12 - 1)]	-0.62

	(1)	(2)	(3)	(4)	(5)
	JPMorgan return, $r_{t,JPM}$		Morgan Stanley, $r_{t,MDT}$		
Month _t		$(r_{t,JPM} - \bar{r}_{JPM}) \div \sigma_{JPM}$		$(r_{t,MS} - \bar{r}_{MS}) \div \sigma_{MS}$	(2) × (4)
Jan	6.8%	$(6.8\% - 3.4\%) \div 6.5\% = 0.523$	7.4%	$(7.4\% - 2.7\%) \div 8.0\% = 0.588$	0.307
Feb	0.8	$(0.8\% - 3.4\%) \div 6.5\% = -0.400$	-0.8	$(-0.8\% - 2.7\%) \div 8.0\% = -0.438$	0.175
Mar	-3.0	$(-3.0\% - 3.4\%) \div 6.5\% = -0.985$	0.5	$(0.5\% - 2.7\%) \div 8.0\% = -0.275$	0.271
Apr	15.4	$(15.4\% - 3.4\%) \div 6.5\% = 1.846$	15.0	$(15.0\% - 2.7\%) \div 8.0\% = 1.538$	2.838
May	-8.7	$(-8.7\% - 3.4\%) \div 6.5\% = -1.862$	-15.7	$(-15.7\% - 2.7\%) \div 8.0\% = -2.300$	4.282
Jun	5.5	$(5.5\% - 3.4\%) \div 6.5\% = 0.323$	7.7	$(7.7\% - 2.7\%) \div 8.0\% = 0.625$	0.202
Jul	4.5	$(4.5\% - 3.4\%) \div 6.5\% = 0.169$	2.5	$(2.5\% - 2.7\%) \div 8.0\% = -0.025$	-0.004
Aug	-5.3	$(-5.3\% - 3.4\%) \div 6.5\% = -1.338$	-6.9	$(-6.9\% - 2.7\%) \div 8.0\% = -1.200$	1.606
Sep	7.1	$(7.1\% - 3.4\%) \div 6.5\% = 0.569$	2.8	$(2.8\% - 2.7\%) \div 8.0\% = 0.013$	0.007
Oct	6.9	$(6.9\% - 3.4\%) \div 6.5\% = 0.538$	8.7	$(8.7\% - 2.7\%) \div 8.0\% = 0.750$	0.404
Nov	5.5	$(5.5\% - 3.4\%) \div 6.5\% = 0.323$	7.4	$(7.4\% - 2.7\%) \div 8.0\% = 0.588$	0.190
Dec	5.8	$(5.8\% - 3.4\%) \div 6.5\% = 0.369$	3.3	$(3.3\% - 2.7\%) \div 8.0\% = 0.075$	0.028
Average return, $\bar{r}$	3.4		2.7		
Standard deviation, σ	6.5		8.0		
				Sum	10.31
				Correlation Coefficient [Sum ÷ (12 - 1)]	0.94

Portfolio Risk and the Importance of Covariance

- Although the portfolio expected return is a straightforward, weighted average of returns on the individual securities, the portfolio standard deviation is *not* the simple, weighted average of individual security standard deviations.

$$\sigma_P = \sqrt{\sum_{j=1}^m \sum_{k=1}^m W_j W_k \sigma_{jk}}$$

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}}$$

Portfolio Risk

You are creating a portfolio of **Stock D** and **Stock BW** (from earlier). You are investing \$2,000 in **Stock BW** and \$3,000 in **Stock D**. Remember that the expected return and standard deviation of **Stock BW** is 9% and 13.15% respectively. The expected return and standard deviation of **Stock D** is 8% and 10.65% respectively. The **correlation coefficient** between BW and D is 0.75.

What is the expected return and standard deviation of the portfolio?

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}}$$

$$r_p = (w_1 \times r_1) + (w_2 \times r_2) + \dots (w_n \times r_n) = \sum_{j=1}^n w_j \times r_j \quad (8.)$$

Where

w_j = percentage of the portfolio's total dollar value invested in asset j

r_j = return on asset j

Portfolio Risk

- $R_p = (3.6\%) + (4.8\%) = 8.4\%$

$$\sigma_p = 0.1091 \text{ or } 10.91\%$$

Portfolio Risk

- Calculation of the standard deviation of a portfolio consisting of 25% JPMorgan and 75% Medtronic, Summarizing what we know about the two stocks already:

Stock	Weight	σ
#1 JPMorgan	0.25	6.5%
#2 Medtronic	0.75	2.8%
Correlation between JPMorgan and Medtronic = -0.62		

Plugging the relevant numbers into Equation 8.9 we have

$$\sigma_p = \sqrt{0.25^2 \times (6.5\%)^2 + 0.75^2 \times (2.8\%)^2 + 2 \times 0.25 \times 0.75 \times 6.5\% \times 2.8\% \times -0.62} = 1.7\%$$

Confirming the result in Example 8.10, the standard deviation of a portfolio that invests 25% in JPMorgan and 75% in Medtronic is just 1.7%, which is less than the standard deviation of either stock on its own.

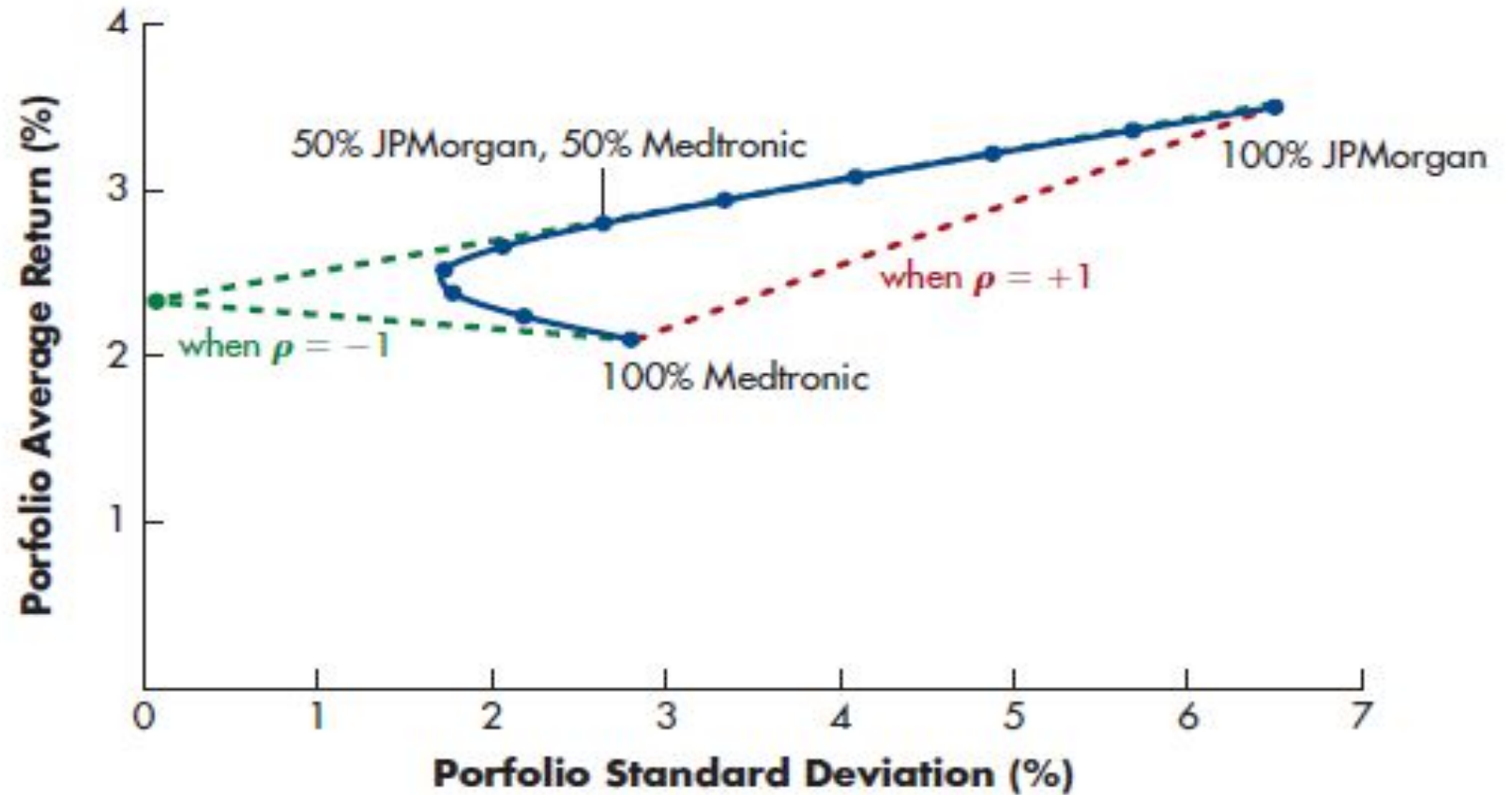
DIVERSIFICATION AND EFFICIENT PORTFOLIOS

- The concept of correlation is essential to developing a portfolio that provides the highest possible return for any level of risk (i.e., an efficient portfolio).
- The concept of diversification makes such common sense that our language even contains everyday expressions that exhort us to diversify (“Don’t put all your eggs in one basket”). The idea is to spread your risk across a number of assets or investments.

DIVERSIFICATION AND EFFICIENT PORTFOLIOS

FIGURE 8.5

Average Return and Standard Deviation of Portfolios of JPMorgan and Medtronic



DIVERSIFICATION AND EFFICIENT PORTFOLIOS

- Any portfolio on that portion of the curve is *inefficient*, meaning that there is another portfolio that invests more in JPMorgan and provides a higher return for the same standard deviation.
- Only portfolios that lie along the portion of the blue arc that slopes up and to the right are *efficient*.
- The low correlation between these two stocks is what causes the arc to bend back and to the left before turning to the right. The red dotted line in Figure 8.5 shows what the average return and standard deviation of portfolios of JPMorgan and Medtronic would look like if the two stocks were perfectly positively correlated. The straight line indicates that, in this special case (when the correlation coefficient is +1), the standard deviation of a portfolio is a simple weighted average of the two stocks' standard deviations.
- The two green dotted line segments in Figure 8.5 illustrates what happens in the opposite extreme case when the correlation between two assets is -1. If JPMorgan and Medtronic were perfectly negatively correlated, portfolios of them would have an average return and a standard deviation lying along one of these line segments. With perfect negative correlation the potential for risk reduction through diversification is at its greatest. In this case, it is always possible to find *some* combination of the two stocks that produces a portfolio that is entirely free of risk, a situation known as hedging.

Total Risk = Systematic Risk + Unsystematic Risk

- Consider what happens to the risk of a portfolio consisting of a single security (asset) to which we add securities randomly selected from, say, the population of all actively traded securities.
- Research studies have looked at what happens to portfolio risk as randomly selected stocks are combined to form equally weighted portfolios. When we begin with a single stock, the risk of the portfolio is the standard deviation of that one stock.
- As the number of randomly selected stocks held in the portfolio is increased, the total risk of the portfolio is reduced. Such a reduction is at a decreasing rate, however. Thus a substantial proportion of the portfolio risk can be eliminated with a relatively moderate amount of diversification – say, 20 to 25 randomly selected stocks in equal-dollar amounts.

Total Risk = Systematic Risk + Unsystematic Risk

Total security risk = Nondiversifiable risk + Diversifiable risk

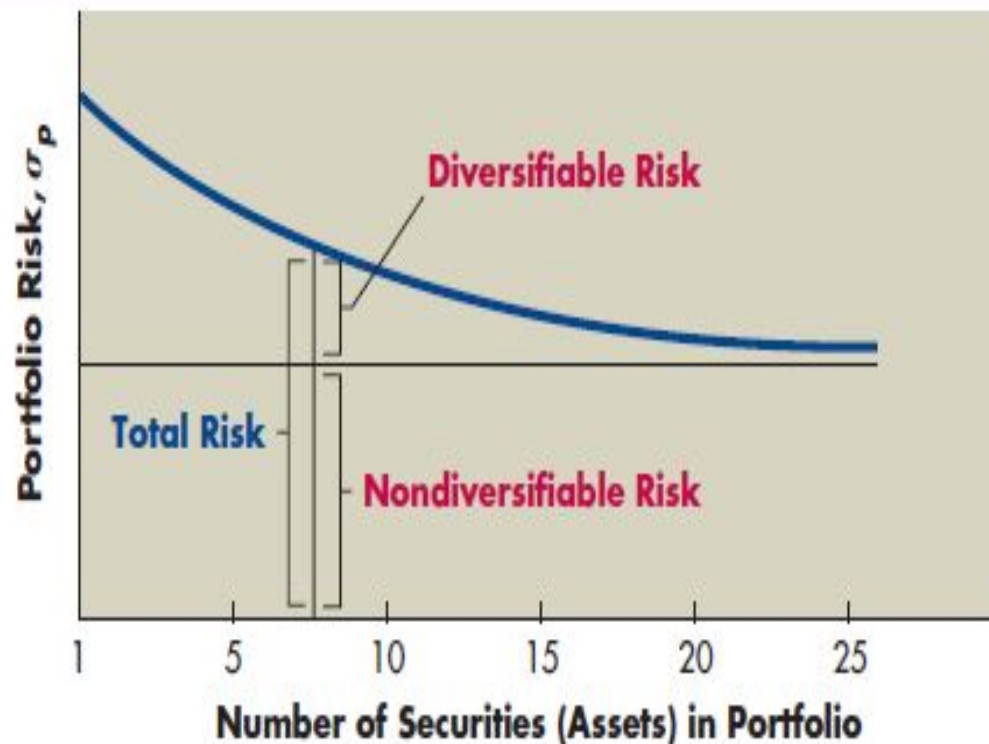
RISK

Systematic risk
Total risk = (nondiversifiable or unavoidable) + Unsystematic risk (diversifiable or avoidable)

- Using the portfolio standard deviation, σ_p , to measure the total portfolio risk

FIGURE 8.7

Risk Reduction
Portfolio risk and diversification



Total Risk = Systematic Risk + Unsystematic Risk

- **Diversifiable risk** (sometimes called unsystematic risk, idiosyncratic risk, unique risk, or firm-specific risk) represents the portion of an asset's risk that is unique to that asset and can be eliminated through diversification.
- It is attributable to firm-specific events, such as strikes, lawsuits, management turnover, or the loss of key accounts. Figure 8.7 shows that diversifiable risk gradually disappears as the number of stocks in the portfolio increases.
- **Nondiversifiable risk** (also called *systematic risk* or *market risk*) is the portion of an asset's risk that is attributable to market factors that affect most if not all firms; diversification cannot eliminate this type of risk because it affects so many firms simultaneously.
- Factors such as war, inflation, pandemics, the macroeconomic cycle, international trade disputes, and political events are examples of nondiversifiable risk. In Figure 8.7, nondiversifiable risk is represented by the horizontal black line below which the blue curve can never go, no matter how diversified the portfolio becomes.

The Capital Asset Pricing Model (CAPM) $V = \frac{D_1}{k_e - g}$

- In market equilibrium, a security is supposed to provide an expected return commensurate with its systematic risk – the risk that cannot be avoided by diversification.
- The greater the systematic risk of a security, the greater the return that investors will expect from the security.
- The relationship between expected return and systematic risk, and the valuation of securities that follows, is the essence of Nobel laureate William Sharpe's **capital-asset pricing model (CAPM)**.
- CAPM is a model that describes the *relationship* between *risk* and expected (required) *return*; in this model, a security's expected (required) return is the **risk-free rate** plus **a premium** based on the **systematic risk** of the security.

CAPM Assumptions

- First, we assume that capital markets are efficient in that investors are well informed, transactions costs are low, there are negligible restrictions on investment, and no investor is large enough to affect the market price of a stock.
- We also assume that investors are in general agreement about the likely performance of individual securities and that their expectations are based on a common holding period, say one year.
- There are two types of investment opportunities with which we will be concerned.
- The first is a risk-free security whose return over the holding period is known with certainty. Frequently, the rate on short- to intermediate-term Treasury securities is used as a surrogate for the risk-free rate.
- The second is the *market portfolio* of common stocks. It is represented by all available common stocks and weighted according to their total aggregate market values outstanding. As the market portfolio is a somewhat unwieldy thing with which to work, most people use a surrogate, such as the **Standard & Poor's 500 Stock Price Index (S&P 500 Index)**. This broad-based, market-value-weighted index reflects the performance of 500 major common stocks.

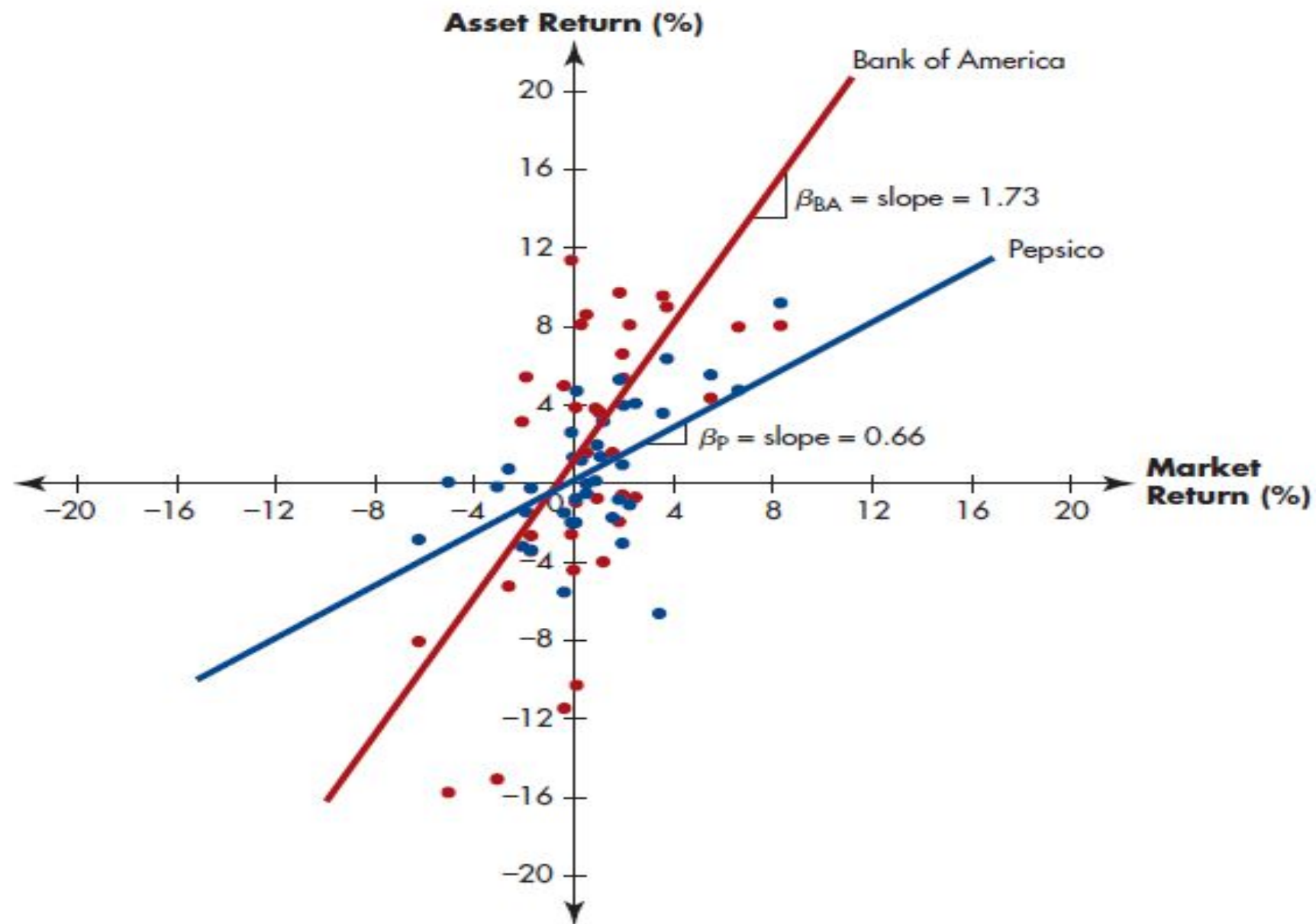
The Characteristic Line

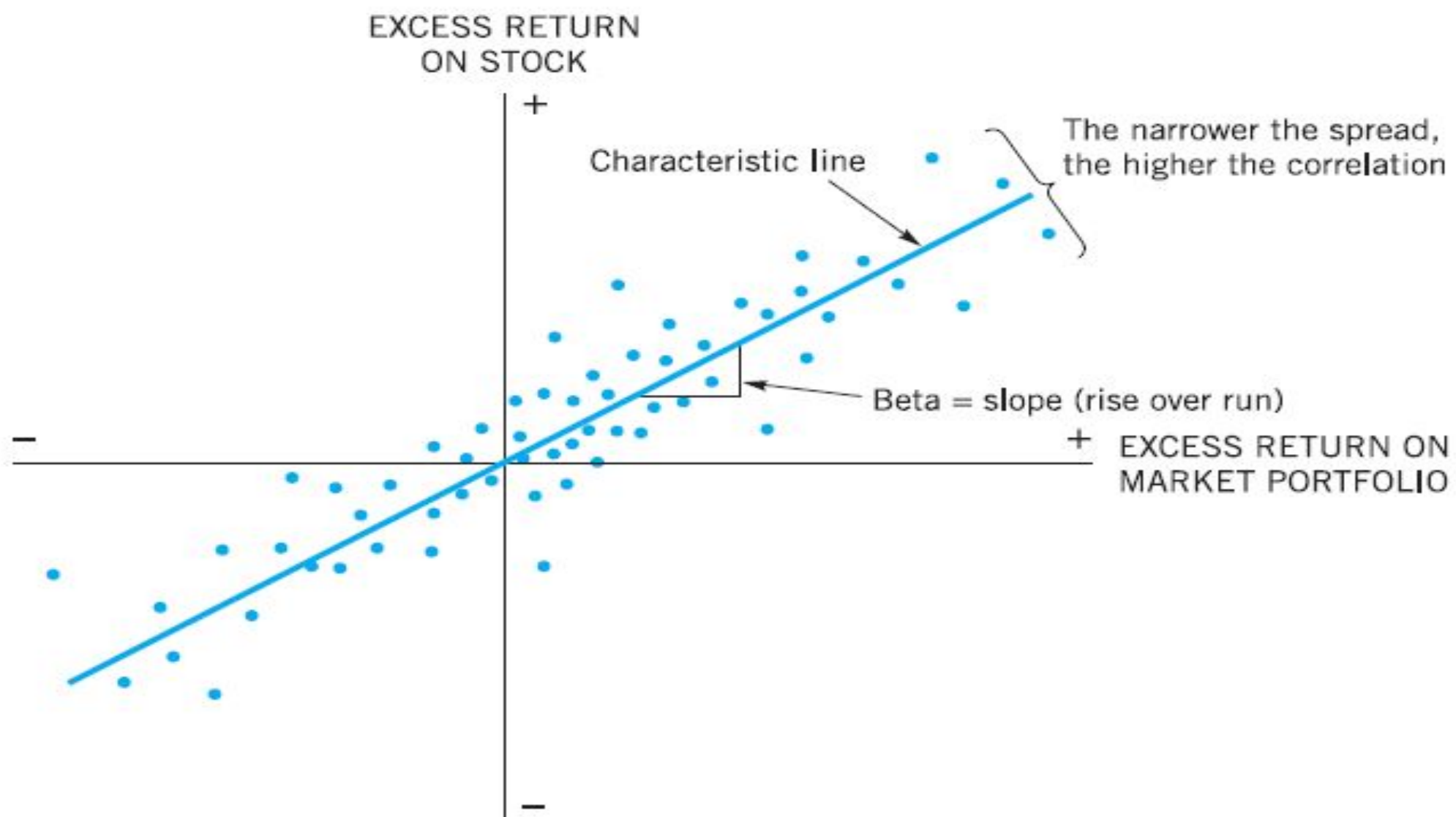
- A line that describes the relationship between an individual security's returns and returns on the market portfolio. The slope of this line is beta.
- Beta coefficient (β) A relative measure of nondiversifiable risk. An index of the degree of movement of an asset's return in response to a change in the market return. (Zutter)
- Beta (β) An index of systematic risk. It measures the sensitivity of a stock's returns to changes in returns on the market portfolio. (VanHorne)
- The beta of a portfolio is simply a weighted average of the individual stock betas in the portfolio.

FIGURE 8.8

Beta Derivation

Graphical derivation
of beta for Bank of
America and
PepsiCo





Interpreting Betas

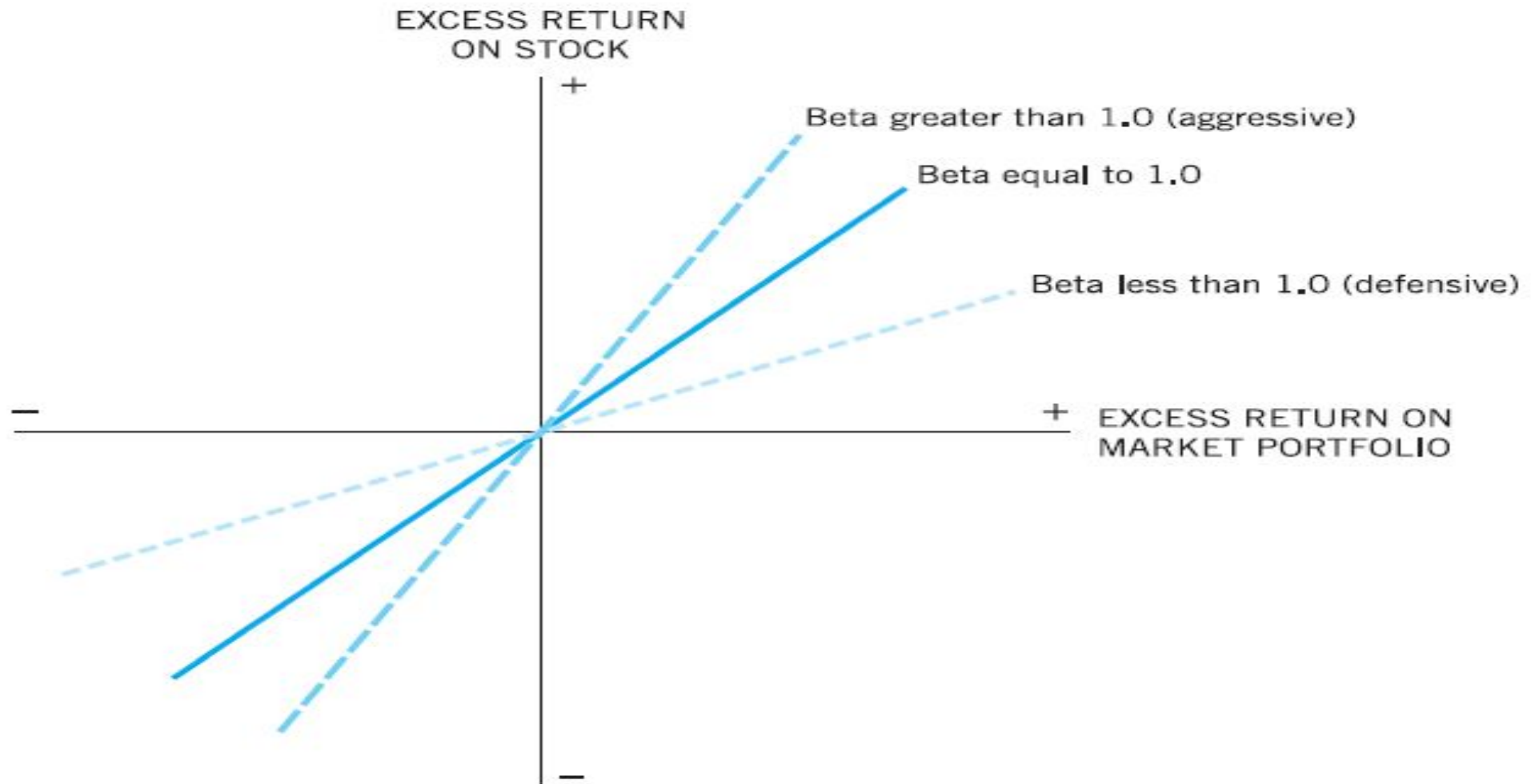
- The beta coefficient for the entire market (and the average beta across all stocks) equals 1.0. All other betas are viewed in relation to this value. Asset betas may be positive or negative, but positive betas are the norm.
- The majority of beta coefficients fall between 0.5 and 2.0.
- The return of a stock that is half as responsive as the market ($\beta = 0.5$) should change (on average) by 0.5% for each 1% change in the return of the market portfolio.
- A stock that is twice as responsive as the market ($\beta = 2.0$) should (on average) experience a 2% change in its return for each 1% change in the return of the market portfolio.

Interpreting Betas

TABLE 8.10**Selected Beta Coefficients and Their Interpretations**

Beta	Direction of response	Magnitude of response
2.0 } 1.0 } 0.5 }	Move in same direction as market	{ Twice as responsive as the market Same response as the market Only half as responsive as the market
0		Unaffected by market movements
-0.5 } -1.0 } -2.0 }	Move in opposite direction to market	{ Only half as responsive as the market Same response as the market Twice as responsive as the market

Interpreting Betas



Interpreting Betas

- Beta is simply the slope (i.e., the change in the excess return on the stock over the change in excess return on the market portfolio) of the characteristic line.
- If the slope is 1.0, it means that excess returns for the stock vary proportionally with excess returns for the market portfolio. In other words, the stock has the same systematic risk as the market as a whole. If the market goes up and provides an excess return of 5 percent for a month, we would expect, on average, the stock's excess return to be 5 percent as well.
- A slope steeper than 1.0 means that the stock's excess return varies more than proportionally with the excess return of the market portfolio. Put another way, it has more unavoidable risk than the market as a whole. This type of stock is often called an "aggressive" investment.
- A slope less than 1.0 means that the stock's excess return varies less than proportionally with the excess return of the market portfolio. This type of stock is often called a "defensive" investment.
- The greater the slope of the characteristic line for a stock, as depicted by its beta, the greater its systematic risk. This means that, for both upward and downward movements in market excess returns, movements in excess returns for the individual stock are greater or less depending on its beta.

Portfolio Beta

- We can easily estimate the beta of a portfolio by using the betas of the individual assets it includes. Letting w_j represent the proportion of the portfolio's total dollar value represented by asset j and letting β_j equal the beta of asset j ,

$$\beta_p = (w_1 \times \beta_1) + (w_2 \times \beta_2) + \dots + (w_n \times \beta_n) = \sum_{j=1}^n w_j \times \beta_j$$

Example

- Mario Austino, an individual investor, wishes to assess the risk of two small portfolios he is considering. The first is a portfolio that Mario called “the necessities,” which includes investments in five stocks listed in Table 8.11: Costco, Exxon, PepsiCo, Starbucks, and Walmart. Mario’s “high-tech” portfolio also contains five stocks: Amazon, Apple, Intel, Microsoft, and Qualcomm. In each portfolio, Mario invests the same proportion (20%)

Stock	Beta	Stock	Beta
Amazon	1.35	Int’l Business Machines	1.24
Apple	1.17	JPMorgan Chase & Co.	1.18
Bank of America	1.73	Microsoft	0.95
Costco	0.70	Netflix	0.97
Disney	1.08	PepsiCo, Inc.	0.66
eBay	1.28	Qualcomm	1.38
ExxonMobil Corp.	1.36	Starbucks	0.80
Gap (The), Inc.	1.43	Verizon	0.44
General Electric	0.96	Walmart	0.32
Intel	0.82	Xerox	1.68

Example

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Bank of America	1.73	Microsoft	0.95
Costco	0.70	Netflix	0.97
Disney	1.08	PepsiCo, Inc.	0.66
eBay	1.28	Qualcomm	1.38
ExxonMobil Corp.	1.36	Starbucks	0.80
Gap (The), Inc.	1.43	Verizon	0.44
General Electric	0.96	Walmart	0.32
Intel	0.82	Xerox	1.68

$$\begin{aligned}\beta_{\text{necessities}} &= (0.20 \times 0.70) + (0.20 \times 1.36) + (0.20 \times 0.66) \\ &\quad + (0.20 \times 0.80) + (0.20 \times 0.32) = 0.77\end{aligned}$$

$$\begin{aligned}\beta_{\text{high-tech}} &= (0.20 \times 1.35) + (0.20 \times 1.17) + (0.20 \times 0.82) \\ &\quad + (0.20 \times 0.95) + (0.20 \times 1.38) = 1.13\end{aligned}$$

CAPM Equation

- Using the beta coefficient to measure non diversifiable risk, the *capital asset pricing model (CAPM)* says

$$r_j = R_F + \beta_j \times (r_m - R_F) \quad (8.12)$$

where

r_j = expected return or required return on asset j

R_F = risk-free rate of return, commonly measured by the return on a U.S. Treasury bill

β_j = beta coefficient or index of nondiversifiable risk for asset j

r_m = expected return on the market portfolio of assets

- Equation 8.12 shows that the CAPM has two parts: (1) the **risk-free rate of return**, R_F , which is the required return on a *risk-free asset*, typically a **U.S. Treasury bill (T-bill)**, a short-term IOU issued by the U.S. Treasury; and (2) the *risk premium*. These parts are, respectively, the two elements on either side of the plus sign in Equation 8.12.
- The $(r_m - R_F)$ portion of the risk premium is called the *market risk premium* because it represents the premium that the investor expects for taking the average amount of risk associated with holding the market portfolio of assets.

CAPM Example

- Mario Austino wants to know what return he should expect from the high-tech portfolio with a beta of 1.13. The risk-free rate of return is 2%, and Mario expects that the risk premium on the market portfolio will be just 4%, much lower than its long-run historical average. This means the expected return on the market portfolio is 6%

$$r = 2\% + 1.13(6\% - 2\%) = 6.5\%$$

The market risk premium of 4% (6% – 2%), when adjusted for the asset's index of risk (beta) of 1.13, results in a risk premium for the high-tech portfolio of 4.5% (6.5% – 2%). That risk premium, when added to the 2% risk-free rate, results in a 6.5% required return.

Other things being equal, the higher the beta, the higher the required return, and the lower the beta, the lower the required return.

The Graph: The Security Market Line (SML)

- The graphical depiction of the capital asset pricing model (CAPM), which shows how the expected or required return in the marketplace depends on non-diversifiable risk (beta).
- The SML is a straight line that shows how an investment's expected return depends on its beta.
- In the graph, we plot non-diversifiable risk as measured by beta, b , on the x -axis, and we plot expected returns, r , on the y -axis. If you know the beta of any asset, you can use the SML to find that asset's expected or required return.

At zero risk, the security market line has an intercept on the vertical axis equal to the risk-free rate. Even when risk is involved, investors still expect to be compensated for the time value of money. As risk increases, the required rate of return increases in the manner depicted.

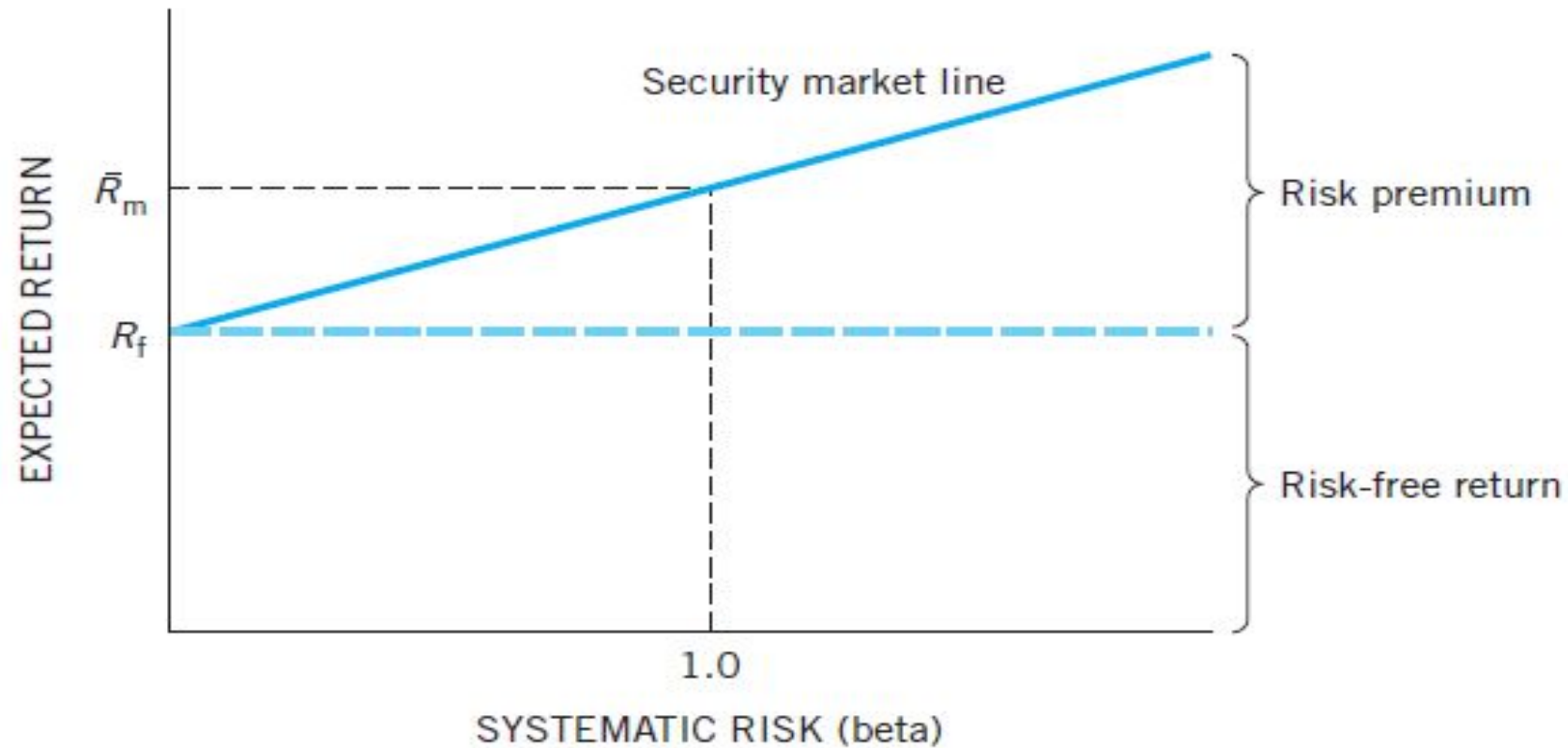
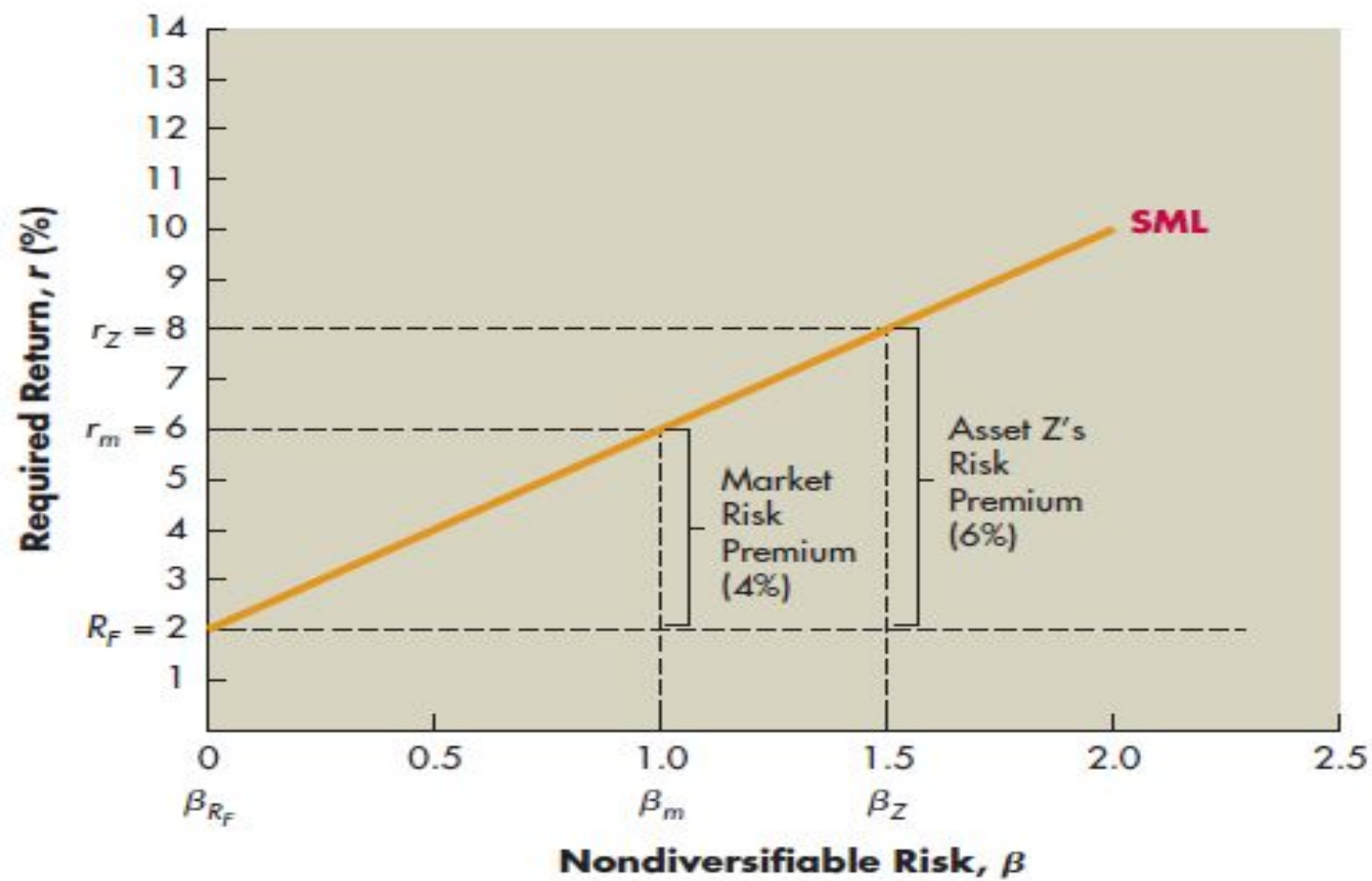


FIGURE 8.9

Security Market Line

Security market line (SML)
with asset Z data shown



Shifts in the Security Market Line

- *Changes in Inflationary Expectations* Changes in inflationary expectations affect the risk-free rate of return, R_F .

$$R_F = r^* + i$$

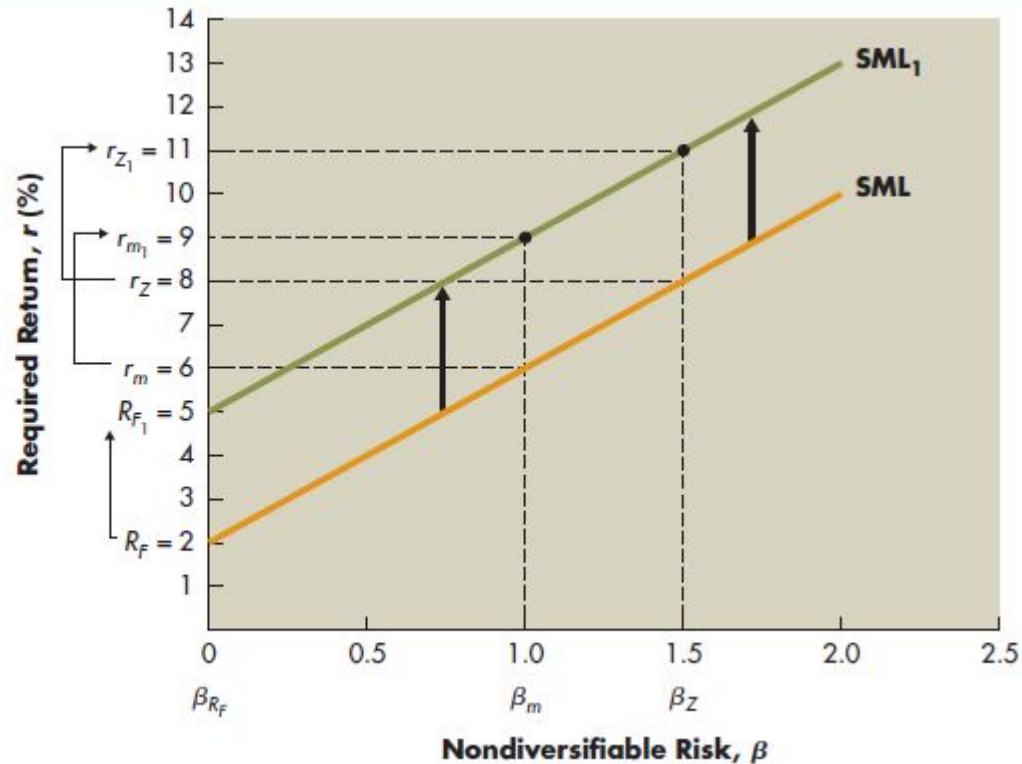
- This equation shows that, assuming a constant real rate of interest, r^* changes in inflationary expectations, reflected in an inflation premium, i , will result in corresponding changes in the risk-free rate.
- Therefore, a change in inflationary expectations that results from events such as international trade disputes or major alterations in Federal Reserve policy will produce a shift in the SML. Because the risk-free rate is a basic component of all rates of return, any change in R_F can affect *all* required rates of return.

Shifts in the Security Market Line

- *Changes in Inflationary Expectations*
- Changes in inflationary expectations result in parallel shifts in the SML in direct response to the magnitude and direction of the change.

FIGURE 8.10

Inflation Shifts SML
Impact of increased inflationary expectations on the SML



Shifts in the Security Market Line

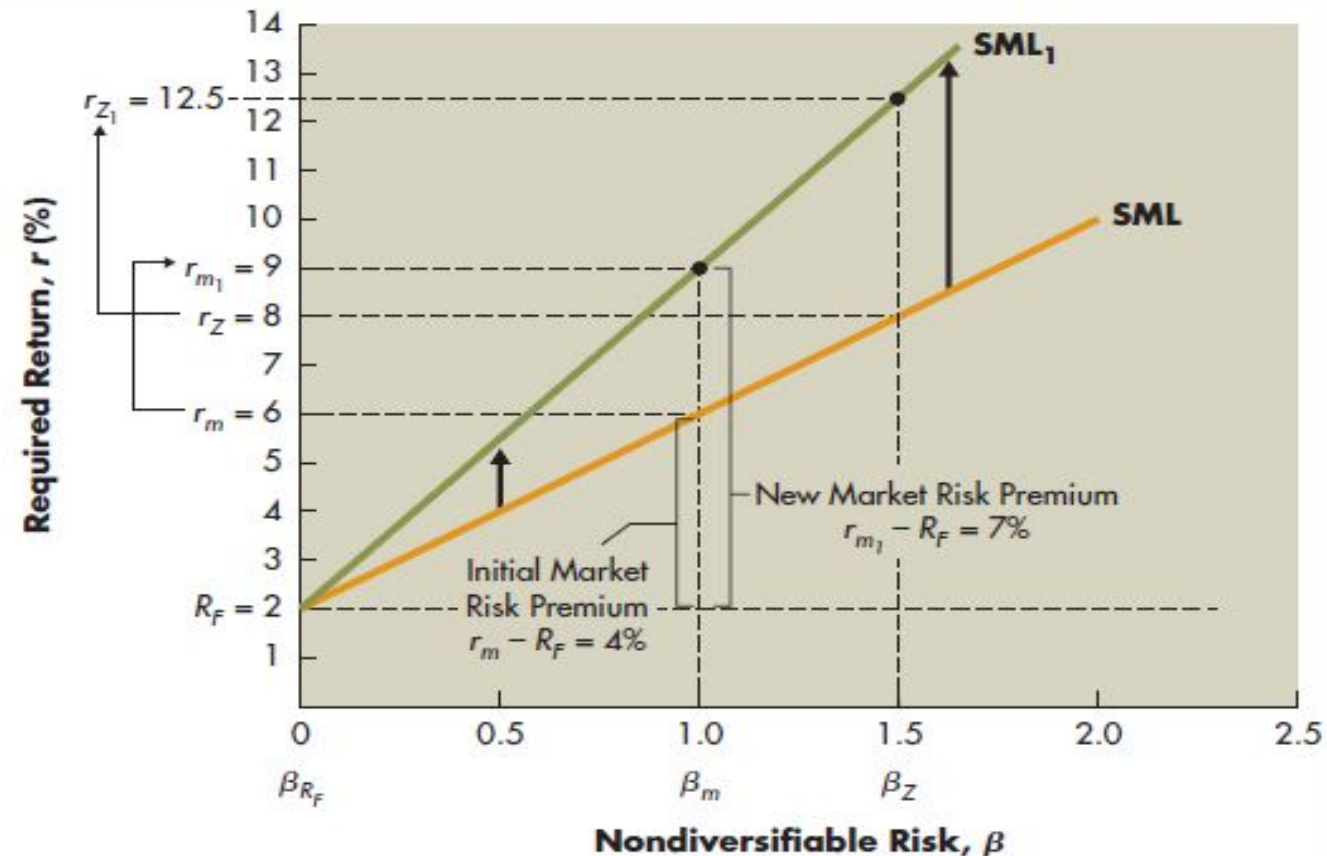
- *Changes in Risk Aversion*
- The slope of the security market line reflects the general risk preferences of investors in the marketplace. As discussed earlier, most investors are *risk averse*, which means they require increased returns for increased risk.
- If investors become more risk averse, then the slope of the SML becomes steeper because when investors are more risk averse, they demand a higher return for any risk level. This also means that risk premiums increase with greater risk aversion.
- Changes in risk aversion and the slope of the SML result from changing preferences of investors, which generally stem from economic, political, or social events.
- Examples of events that *increase* risk aversion include a stock market crash or recession, assassination of a key political leader, and the outbreak of war.

Shifts in the Security Market Line

- *Changes in Risk Aversion*

FIGURE 8.11

Risk Aversion Shifts SML
Impact of increased risk aversion on the SML



Shifts in the Security Market Line

- The security market line is not stable over time, and shifts in the security market line can result in a change in required return. At least two major forces— inflationary expectations and risk aversion—affect the position and slope of the SML.
- *Changes in Inflationary Expectations* Changes in inflationary expectations affect the risk-free rate of return, R_F .

$$R_F = r^* + i$$

Obtaining Betas

- If the past is thought to be a good surrogate for the future, one can use past data on excess returns for the stock and for the market to calculate beta. Several services provide betas on companies whose stocks are actively traded; these betas are usually based on weekly or monthly returns for the past three to five years.
- Services providing beta information include Merrill Lynch, Value Line, Reuters and Ibbotson Associates.
- The obvious advantage is that one can obtain the historical beta for a stock without having to calculate it with their identifying **ticker symbols**, and their stock betas.

Adjusting Beta

- An estimate of a security's future beta that involves modifying the security's historical (measured) beta owing to the assumption that the security's beta has a tendency to move over time toward the average beta for the market or the company's industry.
- There appears to be a tendency for measured betas of individual securities to revert toward the beta of the market portfolio, 1.0, or toward the beta of the industry of which the company is a part. This tendency may be due to economic factors affecting the operations and financing of the firm and perhaps to statistical factors as well.
- If the measured beta was 1.4 and a weight of 0.67 was attached to it and a weight of 0.33 applied to the market beta, the adjusted beta would be $1.4(0.67) + 1.0(0.33) = \mathbf{1.27}$.

Returns and Stock Prices

- The capital-asset pricing model provides us a means by which to estimate the required rate of return on a security. This return can then be used as the discount rate in a dividend valuation model. You will recall that the intrinsic value of a share of stock can be expressed as the present value of the stream of expected future dividends.

$$V = \frac{D_1}{k_e - g}$$

Examples

- Lisa Miller at *Basket Wonders* is attempting to determine the rate of return required by their stock investors. Lisa is using a 6% R_f and a long-term market expected rate of return of 10%. A stock analyst following the firm has calculated that the firm beta is 1.2. What is the *required rate of return* on the stock of *Basket Wonders*?
- Lisa Miller at BW is also attempting to determine the intrinsic value of the stock. She is using the *constant growth model*. Lisa estimates that the dividend next period will be \$0.50 and that BW will grow at a constant rate of 5.8%. The stock is currently selling for \$15. What is the intrinsic value of the stock? Is the stock over or underpriced?

Answer

$$\begin{aligned}\bar{R}_{BW} &= R_f + \beta_j(\bar{R}_M - R_f) \\ \bar{R}_{BW} &= 6\% + 1.2(10\% - 6\%) \\ \bar{R}_{BW} &= 10.8\%\end{aligned}$$

$$\begin{aligned}\text{Intrinsic Value} &= \frac{\$0.50}{10.8\% - 5.8\%} \\ &= \$10\end{aligned}$$

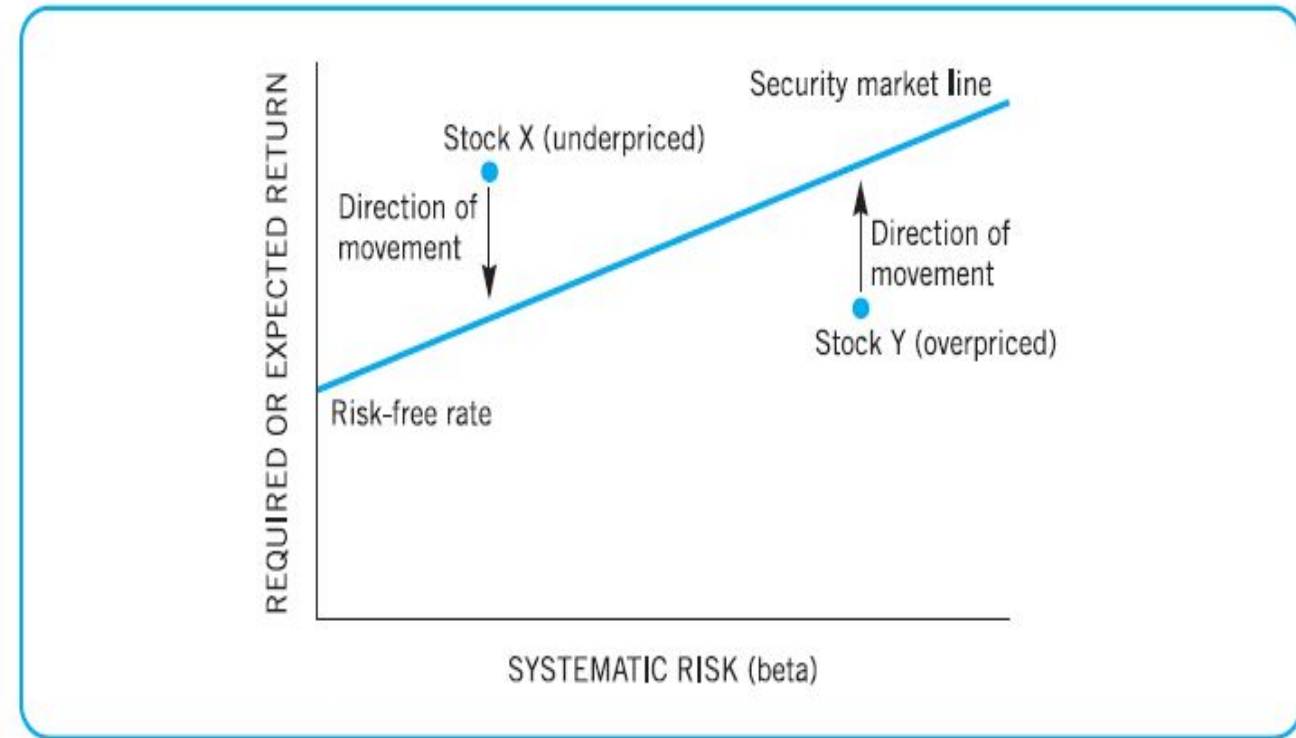
The stock is OVERVALUED as the market price (\$15) exceeds the intrinsic value (\$10).

Example

	BEFORE	AFTER
Risk-free rate, R_f	0.08	0.07
Expected market return, $\bar{R}_m$	0.13	0.11
Savance beta, β_j	1.30	1.20
Savance dividend growth rate, g	0.10	0.09

Underpriced and Overpriced Stocks.

- For some reason, two stocks – call them X and Y – are improperly priced. Stock X is underpriced relative to the security market line, whereas stock Y is overpriced.
- As a result, stock X is expected to provide a rate of return greater than that required, based on its systematic risk. In contrast, stock Y is expected to provide a lower return than that required to compensate for its systematic risk.
- Investors, seeing the opportunity for superior returns by investing in stock X, should rush to buy it. This action would drive the price up and the expected return down.
- In the case of stock Y, investors holding this stock would sell it recognizing that they could obtain a higher return for the same amount of systematic risk with other stocks. This selling pressure would drive Y's market price down and its expected return up until the expected return was on the security market line.



Challenges to the CAPM

- The CAPM has not gone unchallenged. market imperfections as bankruptcy costs, taxes, and institutional restraints have been recognized, and refinements can be made to account for their effects. Some of these refinements are explored in subsequent chapters when we deal with applications of the CAPM.
- When scholars have tried to explain actual security returns, several anomalies (i.e., deviations from what is considered normal) have become evident.
- One is a *small-firm*, or *size*, *effect*. It has been found that common stocks of firms with small market capitalizations (price per share times the number of shares outstanding) provide higher returns than common stocks of firms with high capitalizations, holding other things constant.
- Another irregularity is that common stocks with low *price/earnings* and *market-to-book-value ratios* do better than common stocks with high ratios.
- Still other anomalies exist. For example, holding a common stock from December to January often produces a higher return than is possible for other similar-length periods. This anomaly is known as the *January effect*. Although these January effects have been found for many years, they do not occur every year.
- Other Anomalies are Momentum, liquidity effects, post-earnings drift.

The CAPM and Multifactor Models

- Although the CAPM remains useful for our purposes, it does not give a precise measurement of the market equilibration process or of the required return for a particular stock.
- *Multifactor models* – that is, models which claim that the return on a security is sensitive to movements of multiple factors, or indices, and not just to overall market movements – give added dimension to risk and certainly have more explanatory power than a single-factor model like the CAPM.

Efficient Financial Markets

- An **efficient financial market** exists when security prices reflect all available public information about the economy, about financial markets, and about the specific company involved.
- The implication is that market prices of individual securities adjust very rapidly to new information.
- As a result, security prices are said to fluctuate randomly about their “intrinsic” values. The driving force behind market efficiency is self-interest, as investors seek under- and overvalued securities either to buy or to sell.
- The more market participants and the more rapid the release of information, the more efficient a market should be.

Three Forms of Market Efficiency

- Eugene Fama, a pioneer in efficient markets research, has described three levels of market efficiency:

□ *Weak-form efficiency*: Current prices fully reflect the *historical sequence of prices*. In short, knowing past price patterns will not help you improve your forecast of future prices.

□ *Semi strong-form efficiency*: Current prices fully reflect *all publicly available information*, including such things as annual reports and news items.

□ *Strong-form efficiency*: Current prices fully reflect *all information*, both public and private (i.e., information known only to insiders).

Does Market Efficiency Always Hold?

- Stock market efficiency presents us with a curious paradox: The hypothesis that stock markets are efficient will be true only if a sufficiently large number of investors *disbelieve* its efficiency and behave accordingly.
- In other words, the theory requires that there be a sufficiently large number of market participants who, in their attempts to earn profits, promptly receive and analyze all information that is publicly available concerning companies whose securities they follow.
- We are left with the uneasy feeling that although market efficiency is a good explainer of market behavior most of the time and securities seem to be efficiently priced relative to each other, there are exceptions.
- These exceptions call into question market prices embodying all available information and, therefore, whether they can be completely trusted. Not only are there some extreme events, such as the 1987 stock market crash and 2008 real estate crash, but there are also some seemingly persistent anomalies. Perhaps these anomalies, some of which we discussed earlier, are merely the result of an inadequate measurement of risk.